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References:

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Lecture-1 &2

Human Error and Hazards in the Workplace

Human error

It refers to a mistake made by a person, typically due to judgment, perception, memory, or action. It's when someone does something that was not intended, or fails to do something that was required, which leads to an undesired outcome.

In more detail:

Types of Human Error:

1.Slips and Lapses (Execution errors):

- **Slips**: Doing the wrong thing while intending to do the right thing (e.g., pressing the wrong button).
- **Lapses**: Forgetting to do something due to memory failure (e.g., forgetting a step in a procedure).

2.Mistakes (Planning errors):

- Errors in decision-making or problem-solving—when the plan itself is flawed, even if it's executed correctly (e.g., misinterpreting data).

3.Violations:

- Deliberate deviations from rules or procedures, not always due to jealousy (e.g., skipping steps to save time).

Examples:

- In aviation: A pilot misreads an instrument.
- In healthcare: A nurse gives the wrong dosage of medication.
- In construction: A worker misjudges the load a beam can support

Error models and theories

An **error model** is a structured framework used to understand, analyze, and sometimes predict how and why errors occur—especially **human errors**—in systems, tasks, or processes.

An error model explains the nature, origin, and consequences of errors, typically focusing on how individuals interact with systems and how those systems can fail, either due to human mistakes or design flaws.

1. Swiss Cheese Model (James Reason, 1990)

Type: System-based model

- **Concept:** Think of layers of Swiss cheese stacked together—each layer represents a defence against error. The holes are weaknesses in each defence. When the holes align, an error passes through all layers and leads to an accident.
- **Key takeaway:** Errors often result from a combination of human and systemic failures.
- **Used in:** Aviation, healthcare, engineering, safety-critical systems.



2. Human Error Classification (Reason's Model of Errors)

Reason's model helps in understanding why accidents are often the result of multiple factors, rather than the fault of one individual. By analyzing both the active failures and latent conditions, organizations can implement strategies to improve safety. For example, improving training, designing better systems, or fostering a culture of safety can address the latent conditions that make accidents more likely.

Type: Individual-centered model

- Divides human error into:
  - **Slips:** Execution failures (e.g., hitting the wrong button)
  - **Lapses:** Memory failures (e.g., forgetting a step)
  - **Mistakes:** Planning errors (e.g., misdiagnosing a situation)
  - **Violations:** Intentional rule-breaking

This model helps in diagnosing the type of human error to apply the right corrective action.

3. SHELL Model

Type: Interaction-based model

- **S**oftware (procedures, manuals)
- **H**ardware (tools, equipment)
- **E**nvironment (workplace conditions)
- **L**iveware (human operator)
- Focuses on how the human (Liveware) interacts with each of these components.
- **Used in:** Aviation, ergonomics, and human factors analysis.

4. Skill-Rule-Knowledge (SRK) Framework (Rasmussen, 1983)

Describe how humans perform tasks and make decisions in complex environments. It categorizes human performance into three distinct modes: skill-based behavior, rule-based behavior, and knowledge-based behavior. These modes represent different levels of cognitive processing involved in tasks, depending on the complexity of the situation and the person's experience or familiarity with it.

Type: Cognitive model

- **Skill-based behavior:** Routine actions done without conscious thought.
- **Rule-based behavior:** Following known rules or procedures.
- **Knowledge-based behavior:** Reasoning through unfamiliar problems.

  Error Occurs When: People operate at the wrong cognitive level (e.g., using routine behavior in an unfamiliar situation).

5. Generic Error Modeling System (GEMS)

The Generic Error Modeling System (GEMS) is a framework used for understanding, analyzing, and predicting human errors in complex systems. Developed by James Reason and others, GEMS provides a systematic approach to modeling human error in a way that can help improve system design, safety, and performance. It's widely applied in fields like aviation, healthcare, and other industries where human error can have significant consequences.

- Suggests that different types of errors are associated with different levels of cognitive control:

- Slips and lapses (skill-based)
- Mistakes (rule- or knowledge-based)

🔷 6. Cognitive Load Theory

- Suggests that human working memory has limited capacity, and errors happen when cognitive demands exceed this capacity.
- **Application:** Helps in designing interfaces, procedures, and systems that reduce overload.

Model	Focus	Key Use
Swiss Cheese	System flaws + human error	Safety barriers and accident analysis
SRK	Levels of cognition	Understanding types of error
GEMS	Cognitive error types	Error classification and prevention
SHELL	Human-system interaction	Ergonomics and system design
Cognitive Load	Working memory	Interface and instruction design

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Lecture-3

Error Rates

Error rate refers to the frequency at which errors occur during a task, process, or system operation. It can apply to human performance, machines, or processes.

In Human Factors / Ergonomics:

An **error rate** is usually defined as:

- Example: If a technician makes 2 errors in 100 tasks, the error rate = 2/100 = **2%**

In Engineering or Systems Design:

- Used to **quantify the risk** associated with human involvement in safety-critical tasks (e.g., in nuclear power plants, aviation, healthcare).
- Helps in **designing safer systems**, determining where controls or redundancies are needed.

Error in maintenance task

n maintenance tasks, errors can arise at different stages and for various reasons. Broadly, these errors can be classified into the following types:

1. Human Errors

- **Operational Errors:** Mistakes made by technicians or operators during routine tasks, such as improper handling or incorrect use of tools and equipment.
- **Misdiagnosis:** Incorrectly identifying the cause of a problem, leading to unnecessary repairs or failure to address the actual issue.
- **Communication Errors:** Failure to clearly communicate maintenance instructions, issues, or status updates between team members.
- **Training Deficiencies:** Lack of proper training can result in errors when technicians are unfamiliar with the systems they are maintaining.

2. Design Errors

- **Inadequate Design:** Poorly designed systems or equipment that are difficult to maintain, causing repeated breakdowns or making repairs complex.
- **Lack of Consideration for Maintenance:** When products or systems are designed without considering the ease of repair or parts replacement, leading to errors during maintenance.

- **Poor Documentation:** Incomplete or unclear technical manuals and maintenance schedules can result in incorrect maintenance procedures or overlooked tasks.

3. Process Errors

- **Improper Procedures:** Following incorrect maintenance protocols or skipping steps in the maintenance process, leading to system failures or operational disruptions.
- **Inconsistent Maintenance Scheduling:** Failure to adhere to preventive maintenance schedules, which can cause systems to degrade over time or malfunction.
- **Lack of Standardization:** When maintenance tasks lack uniform procedures across different teams or locations, resulting in variations in performance and error-prone actions.

4. Equipment Errors

- **Component Failure:** A failure of components due to wear and tear, manufacturing defects, or environmental factors, which can be either anticipated or unforeseen during the maintenance process.
- **Inadequate Tools or Equipment:** Using incorrect or outdated tools and equipment during maintenance tasks can lead to improper fixes and further damage.

5. Environmental Errors

- **Unpredictable External Factors:** Factors such as weather, temperature extremes, or power surges that affect the performance of systems or interfere with the completion of maintenance tasks.
- **Hazardous Environments:** Maintenance performed in dangerous or hazardous conditions that increase the likelihood of errors, accidents, or health risks.

6. Systemic Errors

- **Failure of Preventive Measures:** Neglecting to implement or follow through with preventive maintenance practices that would help avoid breakdowns and reduce the need for emergency repairs.
- **Software or System Failures:** Issues in automated systems that are supposed to assist with monitoring, diagnostics, or maintenance scheduling, causing misalignment between scheduled tasks and actual conditions.

7. External Errors

- **Supply Chain Issues:** Delays or shortages in spare parts, tools, or materials needed for maintenance can result in incomplete tasks or subpar repairs.
- **Outsourcing Errors:** If maintenance tasks are outsourced to third parties, lack of oversight or miscommunication can lead to mistakes or missed maintenance work.

8. Software Errors

- **System Bugs or Glitches:** Issues in computerized maintenance management systems (CMMS) or diagnostic software that result in inaccurate data, scheduling mistakes, or improper tracking of maintenance activities.
- **Data Inaccuracy:** Inaccurate data entered into maintenance logs, such as incorrect readings from sensors or equipment, can lead to erroneous maintenance decisions.

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Lecture-4

Avoiding and managing errors

Avoiding and managing errors in maintenance tasks is crucial to ensure smooth operations, enhance equipment reliability, and reduce downtime. Here's how to address and minimize errors:

1. Training and Skill Development

- **Continuous Training:** Regular training programs should be conducted for all maintenance staff to ensure they are up-to-date with the latest equipment, tools, and technologies.
- **Cross-Training:** Cross-training helps technicians understand various systems and equipment, reducing the chance of errors due to unfamiliarity.
- **Simulations and Drills:** Running mock drills or simulations allows staff to practice handling emergency or complex maintenance tasks in a safe environment, improving real-time problem-solving skills.

2. Standardized Maintenance Procedures

- **Create Standard Operating Procedures (SOPs):** Detailed, clear, and standardized maintenance procedures reduce human errors by ensuring consistency across all tasks.
- **Maintenance Checklists:** Provide maintenance teams with checklists that outline the required steps, tools, and safety precautions for each task.
- **Automation of Routine Tasks:** Automating simple, repetitive tasks (e.g., calibration, inspections) using equipment or software reduces human error.

3. Error Reporting and Continuous Improvement

- **Encourage Open Reporting:** Create a culture where errors, no matter how small, are reported immediately without fear of punishment. This helps identify root causes and implement corrective actions.
- **Post-Maintenance Reviews:** After completing major tasks, conduct reviews to assess what went well, what went wrong, and what could be improved. This feedback loop helps refine processes.
- **Root Cause Analysis (RCA):** When errors do occur, use RCA techniques (e.g., the “5 Whys” or Fishbone Diagram) to understand why the error happened and implement solutions to prevent recurrence.

4. Preventive Maintenance (PM)

- **Implement Preventive Maintenance:** Regularly scheduled checks and replacements can reduce the likelihood of unanticipated failures and errors. Preventive maintenance helps identify small issues before they turn into bigger problems.
- **Predictive Maintenance (PdM):** Using sensors and monitoring systems to predict when equipment is likely to fail and addressing potential issues before they cause an error. This reduces downtime and costly repairs.

5. Up-to-Date Documentation and Communication

- **Clear and Accurate Documentation:** Ensure that all systems, maintenance records, and manuals are accurate, up-to-date, and easily accessible to all maintenance staff.
- **Effective Communication:** Foster effective communication between all team members, whether through regular meetings, digital communication tools, or maintenance tracking systems. Clear communication prevents misunderstandings and ensures everyone is on the same page.

6. Leveraging Technology

- **Computerized Maintenance Management Systems (CMMS):** Utilize CMMS to track maintenance schedules, record work orders, and ensure timely actions. This reduces the chance of missed maintenance tasks or overlooked errors.
- **Diagnostic Tools:** Use advanced diagnostic tools and equipment (e.g., sensors, predictive maintenance software) that can detect errors early and provide more accurate diagnoses.
- **Mobile Apps for Maintenance:** Field technicians can use mobile devices to access information, update maintenance logs in real time, and receive instant notifications or troubleshooting advice.

7. Maintenance Audits and Inspections

- **Regular Audits:** Perform regular audits of maintenance procedures to ensure adherence to SOPs and evaluate the effectiveness of the process. This helps catch errors early and improve the maintenance process.
- **Inspections:** Regular inspections of equipment and systems help identify wear and tear or potential failures before they become major issues.

8. Error Management Techniques

- **Fail-Safes and Redundancy:** Implement fail-safe mechanisms and redundant systems to ensure that, if one system fails, there is a backup ready to prevent failure or minimize the impact on operations.
- **Clear Escalation Paths:** Establish clear procedures for escalating errors. If a technician is unsure or unable to solve an issue, they should know exactly whom to contact for assistance.

- **Safety Protocols:** Clear safety protocols ensure that even in cases of errors or accidents, there is minimal risk to personnel and equipment.

9. Collaborative Problem-Solving

- **Team Collaboration:** Maintenance tasks often involve a team effort, and working collaboratively helps reduce errors by providing diverse perspectives and expertise on the problem at hand.
- **Feedback Loops:** Encourage feedback from technicians on the ground regarding challenges they face and ways to improve the maintenance process. Often, frontline workers have valuable insights that can enhance operational procedures.

11. Effective Scheduling

- **Avoid Over-Scheduling:** Ensure that maintenance teams have enough time to complete tasks properly. Rushed work can lead to errors.
- **Consider External Factors:** Schedule maintenance tasks while considering factors like weather, operational load, and equipment usage to avoid working under challenging or unfavorable conditions.

12. Environmental Controls

- **Ensure Proper Working Conditions:** Make sure the work environment (lighting, temperature, cleanliness) is conducive to performing maintenance tasks safely and accurately.
- **Hazardous Environment Training:** Provide specific training for maintenance in hazardous environments to minimize safety risks and operational errors.

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Lecture-5

Implication of error i.e. Accidents

Errors in maintenance tasks can lead to accidents, which have significant implications for both the organization and its employees. These accidents can have serious consequences that affect safety, costs, productivity, and reputation. Below are the key implications of errors leading to accidents:

1. Increased Risk of Injury or Fatalities

- **Physical Injury to Workers:** Errors during maintenance can lead to accidents that cause physical harm, such as burns, cuts, electrocution, falls, or crushing injuries.
- **Fatalities:** In extreme cases, errors can result in fatal accidents. For example, incorrect handling of hazardous materials or failure to properly lock out equipment can lead to catastrophic events.

2. Damage to Equipment and Assets

- **Equipment Breakdown:** Errors during maintenance can damage critical equipment or machinery, leading to operational downtime, loss of productivity, and costly repairs.
- **Long-Term Damage:** Improper maintenance or handling can cause long-term damage that reduces the lifespan of equipment or makes it obsolete, increasing replacement costs.
- **Increased Maintenance Costs:** Mistakes in repairs can require additional labor, parts, or even more complex fixes, thus driving up overall maintenance costs.

3. Financial Losses

- **Emergency Repair Costs:** Accidents often result in the need for urgent repairs, which can be more expensive due to the nature of the damage and the need for specialized labor.
- **Insurance Claims and Legal Costs:** Accidents can lead to insurance claims and potential legal proceedings, particularly if someone is injured. This can result in high costs for medical care, legal settlements, and compensation.
- **Lost Revenue:** Equipment downtime due to maintenance errors can halt production or disrupt services, leading to a significant loss in revenue.
- **Fines and Penalties:** Regulatory bodies may impose fines if the accident resulted from neglecting safety standards or compliance regulations, increasing financial liabilities.

4. Operational Disruption

- **Production Delays:** Maintenance errors that result in accidents may shut down key production processes for extended periods, resulting in delays and an inability to meet deadlines or customer orders.
- **Operational Downtime:** Equipment breakdowns and accidents caused by errors lead to unplanned downtime, which can severely disrupt the overall operations of an organization.
- **Reduced Productivity:** The need to stop work to address accidents, investigate root causes, or handle repairs reduces productivity and affects the efficiency of the maintenance team.

5. Reputation Damage

- **Loss of Trust:** Accidents, especially those involving injuries or fatalities, can damage the organization's reputation with customers, clients, and the public. This loss of trust can harm relationships and deter future business opportunities.
- **Brand Image:** An organization's brand image may suffer, particularly if the accident is widely publicized. This could lead to decreased sales, partnerships, or market share.
- **Negative Publicity:** Media coverage of accidents, especially if caused by preventable maintenance errors, can attract negative attention and public scrutiny, which is difficult to recover from.

6. Legal and Regulatory Consequences

- **Litigation:** If accidents result in employee injury or damage to third-party property, the company may face lawsuits, which can lead to significant legal expenses and reputational harm.
- **Regulatory Non-Compliance:** Accidents may trigger investigations by safety and regulatory authorities and if the company is found to be in violation of safety regulations, they could face further legal penalties, sanctions, or required corrective actions.
- **Workplace Safety Violations:** Persistent errors leading to accidents may prompt audits from regulatory bodies and could lead to stricter safety requirements, impacting the flexibility and operations of the company.

7. Impact on Employee Morale and Engagement

- **Decreased Worker Morale:** Accidents and injuries may demoralize the workforce, especially if workers feel that safety and well-being are not prioritized. This can lead to lower job satisfaction, reduced engagement, and even higher turnover rates.
- **Increased Absenteeism:** Injuries or accidents often lead to medical leave, which reduces the workforce and may cause the remaining workers to take on additional responsibilities, leading to stress and burnout.

- **Loss of Skilled Workers:** If an accident results in the injury or death of a skilled technician or worker, the organization can lose valuable expertise and experience, which can be hard to replace.

8. Environmental Impact

- **Hazardous Material Spills:** If maintenance errors involve hazardous materials or systems, such as chemicals or oil, accidents can lead to spills or leaks that harm the environment. This can lead to cleanup costs and environmental damage that takes years to remediate.
- **Pollution and Contamination:** Errors during maintenance that involve machinery or systems that impact air, water, or soil quality can lead to environmental contamination, regulatory fines, and public backlash.

9. Increased Risk of Further Failures

- **Cascading Failures:** A single maintenance error can trigger a chain reaction, where one malfunction leads to the failure of multiple interconnected systems, causing more widespread disruptions.
- **Compounding Damage:** If an initial error is not caught early, it can lead to more severe failures down the line, creating a situation where multiple repairs or fixes are needed, which exponentially increases costs and complexity.

10. Impact on Safety Culture

- **Erosion of Safety Culture:** Repeated accidents caused by maintenance errors can erode an organization's safety culture. Employees may become complacent or believe that safety measures are not taken seriously, increasing the likelihood of future accidents.
- **Loss of Trust in Management:** If employees feel that management does not take the necessary steps to prevent accidents or address errors promptly, there could be a breakdown in trust between workers and leadership, further deteriorating the safety culture.

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Lecture-6

Managing and Mitigating the Implications of Accidents

To manage and mitigate the implications of accidents caused by maintenance errors, organizations should:

1. **Implement Robust Safety Protocols:** Ensure comprehensive safety protocols are in place, with regular reviews and updates.
2. **Adopt Preventive Maintenance:** Regularly schedule inspections and repairs to prevent sudden failures that could result in accidents.
3. **Provide Comprehensive Training:** Train employees on safety procedures, hazard identification, emergency response, and error prevention.
4. **Conduct Root Cause Analysis (RCA):** After an accident, conduct a thorough RCA to determine the underlying causes and implement corrective actions.
5. **Invest in Technology:** Use predictive maintenance technologies to detect and resolve issues before they result in accidents.
6. **Establish Clear Communication:** Ensure that maintenance teams have clear lines of communication and understand their roles in safety and accident prevention.

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